

Statistics

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Introduction

It is important to understand the underlying statistics to check if the simulation has actually converged or not, and there are multiple methods to see that. This document will focus on a pipeline consisting of several methods to ensure convergence. The document will be updated whenever something else, which is important for the data analysis, is found.

1 Convergence

The following two methods (out of many possible methods) are used to determine whether the data has (almost) converged or not.

1.1 detect_equilibration from pymbar.timeseries^{[1],[2]}

This python method mainly finds the **time needed for equilibration** (t_{eq}), **statistical efficiency** (g), and the effective number of independent samples in the data (N_{eff}). This technique provides the parameters for the key technique, **Block Invariance**, and can be used to tell if the simulation has not converged. The key inferences are:

- t_{eq} : The time after which the time series is almost stationary. If $t_{eq} \geq 0.5 \times \text{len}(\text{data})$, then we won't have enough data for reliable statistics.
- g : This can be used to find the correlation time, the time beyond which two points are almost uncorrelated, as:
$$g = 1 + 2\tau \quad (1)$$
- N_{eff} : The number of independent samples, if $N_{eff} \leq 100$, again, it means that we won't have enough data for reliable statistics (a rule of thumb only).

1.2 Block Invariance

This is the key method which we will use to say that our simulation is having statistically consistent estimates of errors. The procedure is described as follows:

1. Divide the data into blocks of size, like 2τ , 4τ and 8τ .
2. For each block corresponding to every block size, find the mean of every block, and based on that find the SEM (Standard Error of the Mean, $\frac{\text{std}(\text{block means})}{\sqrt{N_{\text{blocks}}}}$, assuming blocks are IID).
3. Use the following expression to judge if different blocks are giving statistically consistent errors or not^[3]:

$$\left| \frac{\text{SEM}_{2\tau} - \text{SEM}_{4\tau}}{\text{SEM}_{2\tau}} \right| \leq \frac{1}{\sqrt{2 \times (N_{4\tau} - 1)}} \quad (2)$$

In the above equation, we have just used 2τ and 4τ , but it can be any two block sizes, $N_{4\tau}$ will be replaced by the number of blocks in case of larger block size. This condition tests whether successive block sizes yield statistically consistent estimates of the standard error.

References

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- [2] J. D. Chodera, W. C. Swope, J. W. Pitera, C. Seok, and K. A. Dill. Use of the weighted histogram analysis method for the analysis of simulated and parallel tempering simulations. JCTC 3(1):26-41, 2007.
- [3] Flyvbjerg, H.; Petersen, H. G. Error Estimates on Averages of Correlated Data. J. Chem. Phys. 1989, 91 (1), 461–466.